k-epsilon equations

Continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.$$

Scalar

$$\frac{\partial \rho C}{\partial t} + \nabla \cdot (\rho C \mathbf{u}) - \nabla \cdot (\mu_T \nabla C) = 0.$$

Momentum

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) + \nabla P - \nabla \cdot \tau = \mathbf{S}_p$$

Energy

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + P)\mathbf{u} - \mathbf{u} \cdot \tau] - \frac{\gamma}{\gamma - 1} \nabla \cdot (\mu_T \nabla T) = S_E$$

Turbulent energy

$$\frac{\partial \rho k}{\partial t} + \nabla \cdot (\rho k \mathbf{u}) - \nabla \cdot (\mu_T \nabla k) = S_k.$$

Turbulent dissipation

$$\frac{\partial \rho \epsilon}{\partial t} + \nabla \cdot (\rho \epsilon \mathbf{u}) - \nabla \cdot (\mu_\epsilon \nabla \epsilon) = S_\epsilon.$$

Turbulent diffusion coefficients

$$\mu_T = \rho C_\mu \frac{k^2}{\epsilon}, \quad \mu_\epsilon = \frac{\mu_T}{1.3}, \quad C_\mu = 0.09.$$

Turbulent production term

$$P_t = \mu_T \left[\frac{\partial u_i}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] - \frac{2}{3} \nabla \cdot \mathbf{u} (\rho k + \mu_T \nabla \cdot \mathbf{u}) \quad \text{(summation convention assumed)}$$

In axisymmetry, the production term is as above with an extra geometric term

$$2\mu_T \frac{u_r^2}{r^2}.$$

The momentum equation source term, $\mathbf{S}_p$, is zero in Cartesians. In axisymmetry it is

$$\mathbf{S}_p = \begin{pmatrix} \mu_T \left[\frac{2}{3r} \nabla \cdot \mathbf{u} - 2\frac{u_r}{r^2} \right] + \frac{1}{r} \left[P + \frac{2}{3r} \rho k \right] \\ \\ 0 \end{pmatrix}$$

The k source term is

$$S_k = P_t - \rho\epsilon.$$

The ϵ source term is

$$S_\epsilon = \frac{\epsilon}{k}(C_1 P_t - C_2 \rho\epsilon), \quad C_1 = 1.4,\ C_2 = 1.94.$$

The turbulent stress tensor, τ, is

$$\tau_{ij} = \mu_T \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \delta_{ij} (\mu_T \nabla \cdot \mathbf{u} + \rho k)$$